

SMART CLASS AI PLANNER

Complete Final Project Report

Course: UYG338 – Software Project Management

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1. Project Identity

Project Name: Smart Class AI Planner

Domain: Educational Software & Time Management

Objective: To develop an intelligent, automated scheduling platform that reduces the manual burden of academic planning while providing data-driven productivity feedback.

2. Abstract

The Smart Class AI Planner is a generative productivity system designed to enhance academic efficiency through automated task management and real-time tracking. Utilizing Python for its core logic, the system implements robust Software Engineering principles such as Encapsulation, Inheritance, and Polymorphism. Key deliverables include a modular backend, a high-fidelity user interface, and an integrated Progress Tracker. The project follows the Agile/Scrum methodology to ensure high-quality delivery and risk-aware development.

3. Introduction

Student life is characterized by complex schedules and multiple deadlines. Traditional planners often fail due to high manual maintenance. This project introduces an "AI Planner" that automates scheduling and motivates users through focus scores and streaks. The scope covers the entire lifecycle from user registration to detailed performance analytics.

4. System Overview

The system operates on a centralized dashboard where users manage study plans. It features a tiered architecture: a secure User/Student layer, a Task orchestration layer, and a polymorphic Reminder engine. The integration of a Progress Tracker allows for continuous monitoring of academic milestones.

5. Technical Foundations

The system is built on an Object-Oriented foundation. High emphasis was placed on modularity, allowing the Project Management team to assign specific modules to different developers without causing integration conflicts. This modularity is achieved through private attributes and controlled interfaces.

6. Project Management (UYG338)

6.1 Methodology

The **Agile/Scrum** framework was utilized. Development was conducted in three major 2-week Sprints, focusing on MVP (Minimum Viable Product) features first, followed by technical depth and UI refinement.

6.2 Task Distribution

Member	Role	Responsibility
Berfin Nazlıcan Öztürk	Project Lead	Architecture & OOP Implementation
Fatma Nur Çetinkaya	UI Designer	User Interface & Dashboard UX
İbrahim Direk	Developer	Backend Logic & Reminder Engine
Damla Daştan	QA Engineer	Testing & Documentation

7. Risk Management

- **Risk:** Logic errors in task completion tracking.
Mitigation: Implementation of automated unit tests for the ProgressTracker class.
- **Risk:** Delivery delays.
Mitigation: Using Jira for real-time progress monitoring and sprint adjustments.

8. Time and Resource Management

Resources were allocated based on member expertise. Time was managed via Gantt charts and weekly stand-up meetings to ensure each work package (Authentication, Scheduling, Analytics) was completed on schedule.

8.1 Project Tracking Tools

- **GitHub:** Used for version control, code reviews, and branch management.
- **Jira:** Used for sprint backlogs, issue tracking, and burndown charts.

9. Special Topic & Technical Depth

The project's technical highlight is the **Automated Progress Analytics**. This module uses an internal algorithm to convert task statuses into quantitative efficiency percentages, providing immediate "Focus Scores" to the user.

10. System Implementation

Implemented in Python, the code prioritizes Clean Code standards. Private attributes (e.g., `__user_id`) ensure data integrity.

```
class Student(User):
    def __init__(self, user_id, name, student_no):
        super().__init__(user_id, name)
        self.__student_no = student_no
        self.__study_plans = []
```

11. Testing and Validation

System testing included functional verification of polymorphic reminder triggers (Email vs. Popup) and stress testing the progress calculation logic with large task datasets. All tests passed with 100% accuracy in status updates.

12. Results and Discussion

The project successfully delivered a robust planning tool. Strengths include the modular OOP architecture and modern UI. A noted weakness is the current lack of a persistent database, which is identified as a primary area for future improvement.

13. Conclusion

The Smart Class AI Planner fulfills the requirements of UYG338 by demonstrating project management maturity and technical excellence. The project provides a solid foundation for a scalable educational productivity platform.